Deepak Kumar

Co-Founder & CTO @ Instruo — EdTech, Ad-Tech, Fintech & AI-Powered Platforms

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Summary

Engineering leader and founder with over 15 years of experience building scalable backend systems ad-server, fintech payment platforms, and AI-driven EdTech products. Currently Co-Founder & CTO at Instruo, an EdTech/LMS platform for hospitality workforce training, where I lead development of AI-generated video course pipelines and micro-learning tools. Deep expertise in Ruby on Rails, microservices architecture, UPI/fintech payment systems, and applied AI (FastAPI, Agentic AI, Temporal workflows). Proven track record scaling systems from monolith to Kubernetes-based microservices across startups and high-growth companies including Truecaller and Chillr.

Experience

Co-Founder & CTO — Instruo (Openvitae)

Apr 2024 – Present (2 yrs 4 mos)

Kochi

- Co-founded Instruo, an EdTech + LMS platform dedicated to hospitality industry training, enhancing workforce skills.
- Developed modules for serving training videos, quizzes, and evaluations to improve the learning experience.
- Built an AI module for generating videos from source content, including full course generation and bite-sized videos.
- Implemented an AI-generated video content pipeline and AI-generated short-form micro-lessons.
- Implemented AI-generated classroom kits including trainer decks, learner workbooks, assessments, and quizzes.
- Implemented Temporal-based workflows to orchestrate long-running tasks for AI services.
- Delivered an experimental virtual training sessions and collaborated cross-functionally to ensure a user-friendly interface and robust platform performance.

Chief Technology Officer — SuperHire (Openvitae)

Aug 2022 – Mar 2024 (1 yr 8 mos)

Kochi

- Spearheaded development of a curated crowdsourcing platform for recruitment sourcing processes.
- Implemented an internal automated filtering mechanism to streamline thousands of candidate evaluation for companies.
- Built and hosted the platform using TypeScript, Hapi.js, and AWS Elastic Beanstalk, later migrated to AWS EKS.

Staff Software Engineer — Truecaller

Feb 2020 – Aug 2022 (2 yrs 7 mos)

Bangalore

- Built an early-stage in-house self-serve Ad-server platform, including CRUD APIs for accounts, orders, campaigns, and creatives.
- Built ad server and ad network integrations for ad serving and fetching ads from third-party partners.
- Optimized ad-serving yield by grouping partners based on targeting and yield parameters to maximize revenue.
- Implemented direct ad serving from partners via fetching and caching third-party ads for direct delivery.
- Developed APIs and server-driven UI rendering for offline ad rendering, and built proprietary ad formats.

Tech: Microservices in Scala & Java, Vue.js, Cassandra, ScyllaDB, Aerospike, Apache Kafka

Senior Software Engineer — Truecaller

Oct 2017 – Feb 2020 (2 yrs 5 mos)

Greater Bengaluru Area

- Built UPI payments for Truecaller's fintech offering, including REST APIs and wrappers for multiple banking APIs.
- Built REST APIs for bill payments and led design & architecture of a microloan product.
- Built a transaction reconciliation system with background jobs for post-payment processing.

- Migrated backend infrastructure from AWS EC2 instances to Kubernetes containers on GCP to enable smooth scaling.

Tech: Microservices in Ruby on Rails, MongoDB, PostgreSQL, Redis, Memcache, RabbitMQ, ELK Stack, Kubernetes on GCP

Senior Software Engineer — Chillr

May 2015 – Sep 2017 (2 yrs 5 mos)

Kochi

- Built REST APIs for UPI payments and developed online merchant payment APIs in a fintech environment.
- Delivered IMPS partner bank integrations and an early version of location-based in-store (QR code) payments.
- Built an expense-splitting feature within the Chillr app, similar to Splitwise.

Tech: Monolith backend in Ruby on Rails, MongoDB, Memcache, RabbitMQ, Splunk, AWS EC2

Earlier Career

- **Software Engineer, 3Dify** (Apr 2014 – Apr 2015) — Joined as first tech employee at a startup building an online marketplace and design studio for 3D-printed products; built CRUD APIs, PayU payment integration, and a from-scratch JS/Handlebars frontend.
- **Software Engineer, WowMakers** (Oct 2012 – Mar 2014) — Built crowdstudio.in, a Ruby on Rails design-crowdsourcing platform similar to 99designs, including onboarding, campaign management, and analytics APIs, deployed on AWS.
- **Junior Software Engineer, Sourcebits, Inc.** (Jun 2011 – Aug 2012) — Built a cross-platform patient admission app with Rhomobile and Rails, and backend APIs for the SkinzGolf golf-tracking application.

Skills

Languages & Frameworks	Ruby on Rails, Ruby, Python, FastAPI, JavaScript, TypeScript, Scala, Java
AI / ML	Agentic AI, Temporal workflow orchestration, AI video & content generation pipelines
Data & Messaging	MongoDB, PostgreSQL, Cassandra, ScyllaDB, Aerospike, Redis, Memcache, Apache Kafka, RabbitMQ
Infrastructure	AWS (EC2, EKS, Elastic Beanstalk), Kubernetes, GCP, ELK Stack, Splunk
Domains	Fintech (UPI, IMPS, bill payments, microloans, reconciliation), Ad-tech, EdTech/LMS

Education

Bachelor of Technology (B.Tech.), Computer Science
University of Kerala — 2007 – 2011 — CGPA – 7.0

Certifications & Languages

B.Tech Degree Certificate — Business English Course, CEFR Level B1
Languages: English (Professional Working), Malayalam (Native/Bilingual), Hindi (Basic)